PwC UK Office Attendance Predictor

Project Proposal

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1. Problem Definition

PwC UK operates a hybrid working model, so daily office attendance varies by site and weekday and cannot be inferred from desk allocations. Facilities Management (FM) uses attendance to plan catering and cleaning, and Real Estate uses it to allocate space. Without forward visibility, FM risks food waste, mis-sized cleaning shifts and shortfalls on busy days.

Events such as bank holidays, rail strikes and internal events also shift attendance. These are logged in a manual form, but the log is historical and not systematically used to forecast attendance, even though many events are known in advance. Public events can be ingested automatically, while internal events will remain a manual input.

The work sits within the Real Estate Data Analytics (REDA) team, whose cloud data platform (Databricks) holds daily headcount totals per office. Forecasting is limited to the built-in forecast in Power BI, a reporting tool, which has not been checked against past attendance or compared with simple rules of thumb, so its accuracy is unknown. Any new model must beat the simple rule of assuming attendance equals the same weekday last week, a standard benchmark in forecasting competitions (Makridakis, Spiliotis and Assimakopoulos, 2020).

2. Project Aim and Objectives

Aim: Develop and evaluate a 1-14 day office attendance forecasting pipeline that outperforms seasonal naive and Power BI baselines, collects public disruption data automatically and provides calibrated prediction intervals.

Objectives:

1. Define KPIs, forecast horizon and simulation assumptions, generate a synthetic dataset mirroring current data structures.
2. Automate collection of bank holidays, strikes and weather warnings into a medallion architecture (raw to analysis-ready layers).
3. Benchmark seasonal naive and Power BI forecasts as baselines.
4. Compare SARIMAX, LightGBM and XGBoost per office using rolling-origin validation (Cerqueira, Torgo and Mozetic, 2020), measured by MAE and MAPE.
5. Produce prediction intervals and report their coverage against the target level.
6. Identify which events most shift attendance using feature importance.
7. Deliver a dashboard for FM and Real Estate, evaluate results and recommend next steps.

Future value: a validated forecast could support wider space optimisation decisions.

3. Research Plan

Prior work will be identified through IEEE Xplore, the ACM Digital Library, ScienceDirect, Google Scholar and arXiv, plus Databricks and MLflow documentation for implementation practice. Searches are mapped to project objectives (Table 1).

Peer-reviewed work from 2018 onwards is prioritised for methods, and from 2020 onwards for hybrid working. Sources are selected for transparent data and methods and relevance to daily or small series. A known limitation is that occupancy estimation is largely sensor-based (Chen, Jiang and Xie, 2018), so findings will guide method choices rather than set expected accuracy. Grey literature, such as real estate and consultancy reports, is labelled and used for context only.

AI tools may support keyword generation and abstract screening, but all sources are read and verified directly. References are managed using mybib. Findings will inform model selection, validation, feature design and interval calibration.

Table 1: Literature search plan by project objective

Objective	Search terms	Sources	Purpose
Simulation	"synthetic time series generation" "hybrid working attendance patterns" "weekday effects office attendance"	Google Scholar, arXiv	Ground simulation assumptions and reduce the risk of unrealistic data
Platform	"medallion architecture" "MLflow experiment tracking"	Databricks and MLflow documentation	Inform pipeline design and experiment tracking
Baselines	"seasonal naive forecasting" "forecast benchmark" "M-competition"	IEEE Xplore, Google Scholar	Confirm baseline choice and standard evaluation metrics
Models	"SARIMAX exogenous variables" "LightGBM time series" "XGBoost forecasting" "rolling-origin evaluation"	ScienceDirect, arXiv	Justify model selection and validation strategy
Events	"calendar effects forecasting" "holiday effect forecasting" "special events demand forecasting" "feature importance time series"	ACM DL, IEEE Xplore	Decide which event features to include and how to encode them
Intervals	"conformal prediction time series" "prediction interval calibration" "quantile regression forecasting"	arXiv, Google Scholar	Choose an interval method suitable for small daily series
Context	"hybrid work office occupancy" "workplace utilisation post-2020"	Google Scholar, grey literature	Ground assumptions about attendance patterns

4. Identification of Technologies, Tools and/or Datasets

Platform: Databricks with Delta tables and MLflow, chosen because the data already resides there and Unity Catalog provides governed access, lineage and experiment tracking. Alternatives are local Python with DuckDB (cheaper, but ungoverned) and Azure ML (heavier infrastructure for a small problem).

Models: Python with statsmodels for SARIMAX (an interpretable classical benchmark) (Hyndman and Athanasopoulos, 2021), plus LightGBM (Ke et al., 2017) and XGBoost (Chen and Guestrin, 2016), which handle exogenous event features flexibly. Prophet is excluded to limit scope, and deep learning because daily office series are small and interpretability matters to FM.

Data: Synthetic data is the primary dataset, generated in Python with Faker and custom simulation code. It mirrors the schema of the REDA attendance, office and date tables, with attendance patterns and event effects simulated from documented assumptions. Bank holidays, Met Office warnings, TfL disruption and strike dates are real public data. No real PwC attendance data is used.

5. Project Plan and Milestones

5.1. Key dates

Table 2: Key project milestones and status

Status	Milestone	Date
Met	Project options shortlisted	10th Oct 2026
Met	Discuss options with manager and lecturer	25th Oct 2026
Met	Finalise choice of project	25th Oct 2026
Planned	Proposal submitted	30 Oct 2026
Planned	Project build and evaluation (16 weeks, assumed)	Start TBC
Planned	Gateway, then project report (6000 words), presentation and KSB mapping	TBC

5.2. Project build schedule (weeks from project start, assumed 16)

Table 3: Project build schedule by week from project start (assumed 16 weeks)

Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
KPIs, horizon and simulation assumptions	■	■														
Synthetic data generator		■	■	■												
Ingestion pipeline (bronze to gold)			■	■	■	■										
Baselines (seasonal naive, Power BI)					■	■										
Model benchmarking in MLflow							■	■	■	■						
Intervals and feature importance											■	■				
Dashboard and benefit estimate													■	■		
Evaluation and recommendations															■	■

Dependencies: modelling depends on the synthetic data generator and ingestion pipeline. The dashboard depends on the chosen model and validated intervals. The report is written after the gateway, outside this schedule.

6. Risk Register

Table 4: Risk register with likelihood, severity and mitigations

ID	Risk	Likelihood	Severity	Mitigation
R1	Synthetic data too simple, so models only recover the generating rules	High	High	Add noise, regime changes and unmodelled events, hold out event types from training, document all generator assumptions in an appendix, state that results show method performance rather than real-world accuracy
R2	Real PwC values or identifiers included by accident	Low	High	Generate from schema only, copy no real values, review all outputs before sharing
R3	Findings do not transfer to real data	Medium	Medium	Frame as a method evaluation, build the pipeline so real data can be swapped in later, base parameters on qualitative patterns described by FM
R4	Public feeds unreliable or change format	Medium	Medium	Store raw responses in bronze, validate on load, fall back to manual flag
R5	Models fail to beat baselines	Medium	Medium	Report as a valid finding, recommend baseline-plus improvements
R6	Scope creep	Medium	Medium	Fixed objectives, MCP and Genie work out of scope
R7	Confidentiality limits on describing PwC systems	Medium	Medium	Agree restrictions with manager on Power BI setup, office counts and schemas, use generic names
R8	Power BI baseline run on synthetic data is only a proxy for the live forecast	Medium	Low	Document the settings used, state the limitation in the evaluation
R9	Compute cost overrun	Low	Low	Small data, job clusters with auto-termination

Ethics and compliance: The project uses synthetic data generated from table schemas only, with no real PwC values and no human participants. QMUL ethics requirements will be confirmed with the module organiser, and confidentiality limits on describing PwC systems will be agreed with the line manager.

7. References

Cerqueira, V., Torgo, L. and Mozetič, I. (2020) 'Evaluating time series forecasting models: an empirical study on performance estimation methods', *Machine Learning*, 109, pp. 1997–2028. Available at: https://doi.org/10.1007/s10994-020-05910-7.

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Makridakis, S., Spiliotis, E. and Assimakopoulos, V. (2020) 'The M4 Competition: 100,000 time series and 61 forecasting methods', *International Journal of Forecasting*, 36(1), pp. 54–74.

8. Appendices

8.1. Appendix A: Generative AI declaration

Generative AI was used in order to help explore potential risks, as well as non-trivial formatting. All AI output was read and verified before being inputted into my project proposal. I take full responsibility for the final content of this document.

8.2. Appendix B: Planned KSB mapping

Table 5: Planned KSB mapping

KSB	Planned evidence
S50	Dashboard and communication of results
S52, S53	Model benchmarking, rolling-origin validation and stability testing
S54, K56	Automated ingestion of public data sources
S55, K54	Analysis of simulated attendance across offices
S2, K3	Risk register and mitigations
S5, S6, K15	Project plan, schedule and time/cost estimation
K18, S14	Research plan and evaluation of forecasting methods
B3	Synthetic data approach, no real PwC data used